

Stocks & Sentiment: Investigating Whether WallStreetBets Sentiment Predicts Short-Term Price Movements

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1. Introduction & Problem Statement

Online communities increasingly discuss and even move financial markets. In particular, the subreddit r/WallStreetBets (WSB) has become famous for highly speculative stock and options trading, and in some cases has been blamed or credited for sharp price movements (ex, the GameStop short squeeze). This raises an important question for both investors and researchers:

Does sentiment on social media about a company predict its short-term stock price movements, or is it mostly a reaction to what has already happened in the market?

This project will investigate that question using Reddit posts from r/WallStreetBets combined with historical stock price data. The core idea is to measure sentiment about specific tickers on a given day and examine how that sentiment relates to next-day price changes (ex, returns, “up vs down” labels) for those stocks. The outcome will contribute to understanding whether WSB sentiment acts more like a leading indicator (influence) or a lagging indicator (reaction), and will demonstrate a machine-learning pipeline that combines NLP sentiment analysis with financial time-series modeling.

2. Data Sources

Social Media / Sentiment Dataset

- **Reddit WallStreetBets Posts (Kaggle)**
 - URL: <https://www.kaggle.com/datasets/gpreda/reddit-wallstreetsbets-posts>
 - Description: This dataset contains posts collected from the r/WallStreetBets subreddit, including fields:
 - title, selftext (post text),
 - created_utc (timestamp),
 - score, num_comments,
 - and additional metadata (ex, author, id).
 - Relevance: r/WallStreetBets is a trading-focused community where users frequently mention specific stock tickers and express strong opinions (ex, “GME to the moon”, “sell everything”). It is a natural environment for capturing retail sentiment toward individual companies.

Financial / Price Data

- **Yahoo Finance market data, accessed via the yfinance Python library**
 - yfinance is an open-source Python library that provides a convenient interface to historical price data (open, high, low, close, volume) and other financial information from Yahoo Finance.
 - It supports downloading daily historical data for multiple tickers within a specified date range and returns the results directly as pandas DataFrames.

Data Linking & Time Alignment

- Convert created_utc timestamps in the WSB data to dates (in a consistent timezone, e.g., UTC or U.S. Eastern).
- For each ticker and date:
 - Aggregate sentiment features from posts on that date.
 - Then align them with price data for that ticker on the following trading day.
- This structure explicitly distinguishes posts as potentially predictive signals (prior-day sentiment) rather than purely reactions to same-day price movements.
- I may also compute and compare “reaction” features by looking at posts on the same day as price changes to see which is more strongly correlated with returns.

3. Methods, Techniques & Technologies

Preprocessing & Feature Engineering

1. Ticker Extraction & Filtering

- Use regular expressions and/or a predefined list of U.S. stock tickers to extract ticker symbols from each post's title and selftext.
- Filter out posts that do not contain any valid ticker symbols.
- Select the top N tickers by number of posts (ex., 5–10), such as GME, AMC, TSLA, AAPL, etc., to ensure adequate sample size per ticker.

2. Text Cleaning & Sentiment Scoring

- Clean the text by lower-casing, removing URLs, user mentions, and non-informative tokens.
- Compute post-level sentiment using: a lexicon-based method (ex, VADER) to get a sentiment polarity score (negative/neutral/positive) and compound score.
- For each ticker and date, aggregate sentiment into daily features such as:
 - Average sentiment score
 - Proportion of positive vs negative posts
 - Volume of posts (num_posts), total/upvote-weighted sentiment

3. Price-Based Features

- From Yahoo Finance price data, compute:
 - Daily return $((\text{Close}_t - \text{Close}_{t-1}) / \text{Close}_{t-1})$
 - Rolling averages (e.g., 5-day and 10-day moving average returns or prices)
 - Volatility proxies (e.g., rolling standard deviation of returns)
- Label each (ticker, day) instance with next-day movement:
 - Regression target: NextDayReturn (continuous).
 - Classification target: NextDayUpDown (1 if return > 0, else 0).

4. Influence vs Reaction Design

- Predictive (influence) setting: use sentiment from day t to predict price movement on day $t+1$.
- Reaction setting: examine how sentiment on day t correlates with the same day's price movement.
- Comparing these two settings will help answer whether WSB sentiment tends to lead the market (influence) or follow it (reaction).

Modeling

1. Baseline Models

- **Price-only model:**
 - Logistic regression, Random Forest, or XGBoost using only price-based features (past returns, volatility, moving averages) to predict NextDayUpDown.
 - Serves as a benchmark for how far traditional price information goes without sentiment.

2. Sentiment-Only Models

- Models trained with only sentiment features (average sentiment, positive/negative ratios, post counts) to predict NextDayUpDown.
- Explore whether sentiment alone has any predictive power across different tickers.

3. Combined Models (Sentiment + Price)

- Gradient-boosted trees (e.g., XGBoost or LightGBM) or Random Forests using both:
 - sentiment features, and
 - price-based features.
- Compare performance against baseline models to test whether sentiment adds incremental predictive value.

4. Time-Aware / Cross-Validation Strategy

- Use time-series-aware train/test splits (ex, train on the first ~70% of dates and test on the remaining ~30%) to avoid look-ahead bias.
- If time, perform rolling-window evaluation to see how model performance changes across different market periods (ex, calm vs volatile periods).

4. Primary Deliverables

- Git Repository

- Use of AI Document
- Notebook(s)
- Final Report
- Short Video Overview
- Model Artifacts (Optional Bonus)

5. References

- G. Preda, “Reddit WallStreetBets Posts” dataset, Kaggle.
- yfinance Python library and documentation (historical market data from Yahoo Finance)
- Relevant literature on social media sentiment and financial markets (ex, papers studying Twitter or Reddit sentiment and stock returns; will be expanded in the final report’s References section).